

COLUMBIA, CA, USA

WMO: 720615

Lat: 38.033N

Lon: 120.417W

Elev: 2120

StdP: 13.60

Time Zone: -8.00 (NAP)

Period: 09-19

WBAN: 00206

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	26.7	27.9	15.6	13.1	47.3	18.8	15.3	50.9	12.5	47.0	10.9	48.0	0.2	50	0.446

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	35.2	100.2	65.1	98.7	64.5	95.0	63.1	68.1	92.9	66.5	91.3	65.3	90.2	6.3	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.3	81.5	72.6	57.4	75.9	71.3	56.5	73.6	70.6	33.4	92.9	32.2	90.3	31.1	90.0	72.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.9	9.7	8.1	DB	21.8	104.2	3.2	2.6	19.5	106.1	17.6	107.6	15.7	109.1	13.4	111.0	(4)
(5)			WB	20.2	69.0	2.7	2.1	18.3	70.5	16.7	71.7	15.2	72.9	13.2	74.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.9	46.3	46.8	50.1	54.2	60.5	70.3	76.5	75.8	70.8	60.4	50.2	44.1		
	DBStd	13.06	5.45	5.95	6.33	7.06	7.16	7.57	5.01	4.61	6.96	6.34	6.59	5.81		
	HDD50	653	137	121	79	37	5	0	0	0	0	3	77	194		
	HDD65	3388	578	510	463	328	175	35	0	2	31	173	445	648		
	CDD50	3904	23	31	81	165	332	608	822	801	623	324	82	12		
	CDD65	1162	0	0	0	5	37	193	357	337	204	29	0	0		
	CDH74	19138	0	1	31	224	896	3386	5703	5092	3079	684	42	0		
	CDH80	10538	0	0	3	45	305	1841	3470	2992	1649	231	2	0		
(14) Wind	WSAvg	2.4	1.6	2.4	2.7	2.8	2.9	2.9	2.8	2.7	2.3	1.9	1.6	1.7		
(15) Precipitation	PrecAvg	4.5	8.9	7.4	5.9	3.7	2.6	0.7	0.1	0.1	0.4	2.6	4.1	8.3		
	PrecMax	70.5	21.9	17.9	19.3	11.6	6.5	2.0	1.7	1.2	1.9	8.3	9.5	19.6		
	PrecMin	24.8	0.7	0.4	0.3	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.4	0.9		
	PrecStd	13.3	6.3	4.7	5.1	3.0	1.9	0.7	0.4	0.2	0.5	2.4	2.7	5.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.5	72.5	78.8	84.3	91.3	102.4	103.9	102.4	99.7	91.2	78.8	66.4		
		MCWB	50.5	55.0	57.6	62.1	62.7	67.8	65.4	65.2	63.6	60.4	55.2	48.8		
	2%	DB	66.0	68.3	72.4	81.0	86.4	99.4	100.2	99.5	95.4	86.0	74.6	63.2		
		MCWB	49.0	53.3	56.6	60.6	61.2	66.0	65.0	64.0	61.5	59.1	54.6	49.8		
	5%	DB	63.2	64.4	68.4	75.3	82.5	94.7	98.8	95.4	93.0	81.5	70.3	58.9		
		MCWB	49.0	52.0	54.0	57.6	59.6	64.1	64.7	63.0	61.6	57.3	53.5	48.7		
	10%	DB	58.9	60.9	64.4	72.0	79.3	90.5	94.9	93.1	90.3	76.9	64.0	55.1		
		MCWB	48.2	50.1	52.6	56.7	57.8	62.3	63.6	62.3	61.2	55.9	51.5	48.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.3	58.5	59.9	64.6	65.5	70.4	69.5	68.0	67.4	64.1	60.8	57.2		
		MCDB	62.4	65.4	73.9	81.2	87.5	96.8	96.4	94.8	90.7	79.8	72.3	60.5		
	2%	WB	53.5	55.8	57.6	61.3	62.9	68.3	67.9	66.3	64.8	62.2	57.4	54.4		
		MCDB	58.6	64.0	69.9	77.2	83.2	92.6	93.3	93.7	89.5	79.6	70.4	56.5		
	5%	WB	51.6	53.6	55.6	59.1	61.1	66.4	66.4	65.1	63.2	59.9	55.0	52.1		
		MCDB	58.2	61.9	66.2	73.8	79.3	89.4	92.1	92.3	88.3	76.6	65.7	55.2		
	10%	WB	49.7	51.1	53.6	57.1	59.5	64.4	65.0	63.9	61.9	57.7	52.6	49.7		
		MCDB	56.5	58.9	62.4	69.5	77.2	86.2	91.1	90.1	86.9	71.8	61.1	54.1		
(35) Mean Daily Temperature Range	MDBR		22.8	22.5	23.0	26.2	28.5	33.5	35.2	34.5	32.9	28.8	24.4	20.7		
	5% DB	MCDBR	29.2	29.9	30.2	32.7	34.8	37.8	37.5	36.5	36.7	34.4	31.7	25.9		
		MCWBR	15.9	16.6	16.0	16.1	13.9	12.5	11.5	12.0	13.1	13.9	15.6	15.0		
	5% WB	MCDBR	21.6	26.5	26.9	30.7	32.3	34.7	34.2	34.0	33.4	29.7	27.7	17.9		
		MCWBR	13.8	16.1	15.2	16.0	13.7	12.0	11.0	11.4	12.5	13.0	15.7	13.4		
(40) Clear-Sky Solar Irradiance	taub		0.320	0.318	0.341	0.336	0.348	0.340	0.350	0.332	0.335	0.318	0.314	0.298		
	taud		2.427	2.460	2.388	2.432	2.422	2.459	2.451	2.485	2.483	2.530	2.491	2.520		
	Ebn at Noon		272	290	292	299	295	296	293	297	291	286	273	274		
	Edh at Noon		27	30	35	36	36	35	35	33	31	27	25	23		
(44) All-Sky Solar Radiation	RadAvg		736	1036	1476	1918	2325	2648	2603	2358	1921	1379	903	657		
	RadStd		113	136	164	177	157	114	93	74	69	93	62	103		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)	+0.74	N/A	-1.98	N/A	N/A	N/A	N/A	-209	+250	+76

Nomenclature: See separate page